Research and Design

Introduction

Learning Objectives

By the end of this lesson, BSIT students should be able to:

✔ Define research and explain its purpose in IT/IS
✔ Identify characteristics of research
✔ Differentiate types of research with examples
✔ Explain research methods in IT studies
✔ Define design and compare it with research
✔ Distinguish the gray area between design & research
✔ Differentiate research vs. design problems

Meaning of Research

Definition: A systematic investigation to refine techniques of thinking, employing specialized tools, instruments, and procedures to obtain a more adequate solution to a problem than would be possible under ordinary means.

- **According to the Oxford Concise Dictionary** defines research is defined as a systematic investigation into and study of materials, sources, etc. to establish facts and reach new conclusions.
- It is an endeavor to discover new or collate old facts by the scientific study of a subject through a course of critical investigation.

💡 Example: Studying the effectiveness of a mobile app for online attendance.

Purpose of Research

Definition: Research aims to explore, describe, explain, and predict phenomena. It also serves man, and the goal of research is a good life, to satisfy man's craving for more understanding, to improve his judgments, to add to his power, to reduce the burden of work, to relieve suffering, and to increase the satisfaction in multitudinous ways – these are the large fundamental goals of research.

🎯 **Purposes:**

- Provide evidence for IT solutions
- Understand user/organizational needs
- Validate project feasibility and impact

Characteristics of Research

Definition: Essential qualities that make research reliable and valid.

⚙️ Systematic – structured approach (SDLC, Agile)

📏 Logical – based on IT principles

📊 Empirical – based on data/testing

🔁 Replicable – can be repeated/tested

🔍 Analytical – uses frameworks, tools, algorithms

Types of Research (with Examples)

Definition: Approaches researchers use depending on their goals.

🧪 Applied Research → Builds IT solutions (e.g., e-commerce system)
📋 Descriptive Research → User surveys on system usability
⚡ Experimental Research → Testing algorithms for efficiency
🗣️ Qualitative Research → Interviews on user experience
📈 Quantitative Research → Measuring system performance (response time)

Research Methods in IT

Definition: Strategies used to collect and analyze data.

📂 Case Study → Analysis of an existing IT system
📝 Survey Method → Gathering user requirements
🧪 Experimental → Testing database queries/system modules
💻 Prototyping → Creating demo systems for validation

Meaning of Design

Definition: A creative and systematic process of planning solutions in IT.

- 💡 Includes UI/UX design, database schema, and system architecture.

Research and Design Distinguished

🔍 Research → Understands the problem (e.g., what causes data loss in hospitals?)

⚙️ Design → Creates solutions (e.g., design a secure hospital records system)

The Gray Between Research & Design

Definition: The overlap where research informs design, and design validates research.

Example: In Capstone, students research user needs before designing the system.

Research vs. Design Problem

🔎 Research Problem → What factors affect online students’ low participation?

⚙️ Design Problem → How can we design a web platform to increase participation?

Class Activity (Capstone-Oriented)

Scenario: Local businesses struggle with inventory management.

👩‍🔬 Researchers → Frame a research problem (e.g., challenges of manual tracking)

👨‍💻 Designers → Frame a design problem (e.g., web-based inventory system)

📢 Present research vs. design perspective in class

Reflection Question

- 💭 **Why is it important for BSIT students to balance research and design in their Capstone Project?**

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